Stable Supply Chains, Falling Growth

Xiao Sun

London School of Economics

Literature Review

The Decline of Business Dynamism and Aggregate Growth

- **Less dynamism.** Fewer firms enter, fewer jobs are created and destroyed each year, fewer young firms grow fast.
- **More concentration.** Sales concentrate in the largest firms, markups rise, the labour share falls.
- **Less growth.** Productivity growth slows after 2004 while research effort rises.

Literature.

- Slower labour-force growth (Karahan, Pugsley & Şahin 2024; Hopenhayn, Neira & Singhania 2022; Peters & Walsh 2026)
- Superstar firms and intangibles (Autor et al. 2020; Hsieh & Rossi-Hansberg 2023; Aghion et al. 2023; De Ridder 2024)
- Less knowledge diffusion (Akcigit & Ates 2021, 2023)
- Low interest rates (Liu, Mian & Sufi 2022)
- Harder innovation (Bloom et al. 2020; Olmstead-Rumsey forthcoming; Cavenaile, Celik & Tian forthcoming)
- Frictions and regulation (Decker et al. 2020; Davis & Haltiwanger 2014; Grullon, Larkin & Michaely 2019)

▸ Appendix: numbers and sources

Mechanism and Motivating Facts

This Paper

The role of the supply chain on business dynamism and aggregate productivity growth post 2000

A new link/technology delivers a more volatile supply. As time goes by, the volatility decreases. Since 2000, this volatility difference has been increasing. ▸ Fact 1

↓

In the supply chain, buyers value the stability of their inputs. ▸ Fact 2

↓

Buyers have the tendency to keep mature links/productivity. ▸ Fact 3

↓

Mature producers/market leaders have less incentive to innovate, as new technology comes with high volatility in the first place. ▸ Fact 4

↓

With fewer takeovers and fewer leader innovations, markups rise and growth slows.

Fact 1: Mature Links Deliver a More Stable Supply, and Increasingly So Since 2000

ratio of SDs of sales growth

	2001–05	2006–10	2011–15	2016–20
age 1 / age 2	1.12	1.09	1.22	1.24
age 1 / age 3	1.19	1.19	1.33	1.32
age 1 / age 4	1.25	1.25	1.33	1.32
age 1 / age 5	1.32	1.30	1.42	1.36
age 1 / age 6	1.37	1.37	1.46	1.70
age 1 / age 7	1.37	1.45	1.55	1.81
age 1 / age 8+	1.56	1.62	1.76	1.86

Compustat segment customers, all links, 2001–2020.

◂ Back to This Paper ▸ Appendix: 1981–2024

Fact 2: Links Last Longer at Large, Stable Suppliers

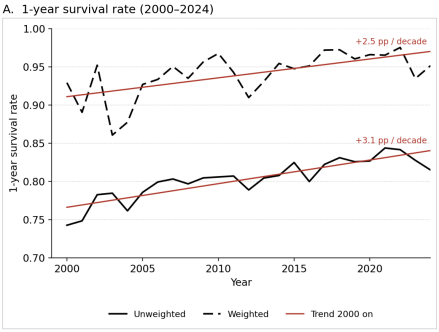
$$\Pr\big(\text{link } sc \text{ alive in } t{+}1 \mid \text{alive in } t\big) = \alpha_{\text{tenure}_{sct}} + \beta^{\top} x_{s,t-1} + \gamma_t + \delta_{\text{SIC}_s} + \varepsilon_{sct}$$

	2000–2024			
	(1)	(2)	(3)	(4)
log employment$_{t-1}$	$+2.46^{***}$			$+1.72^{***}$
	(0.31)			(0.32)
log(1+R&D per worker)$_{t-1}$		-0.58		$+0.26$
		(0.36)		(0.37)
SD of sales growth, $t-5$ to $t-1$			-2.67^{***}	-2.17^{***}
			(0.29)	(0.31)
link-years		67,228		
mean annual survival		0.814		

pp per SD of the lagged supplier characteristic on the annual survival rate; tenure, year and supplier SIC2 fixed effects; standard errors clustered by supplier. 2000–2024 pooled.

◂ Back to This Paper ▸ Appendix: deeper lags

Fact 3: B2B Links Have Lasted Longer Since 2000





US Compustat, links between listed suppliers and their major customers. Survival rate = share of links in year t still reported in $t+1$ (panel A) or in every year through $t+5$ (panel B). Weighted: $S_t^w = \sum_i \left(s_{it} / \sum_j s_{jt} \right) a_{it}$, with s_{it} link i's sales in t and $a_{it} = 1$ if it survives. Red line: least-squares trend, slope in percentage points per decade.

◂ Back to This Paper | ▸ Appendix: Belgium, all B2B links | ▸ Appendix: full sample 1978–2024 | ▸ Appendix: by supplier age

Fact 4: Suppliers with Longer Link Duration in the Past Are Now Large in Sales and Light in R&D Intensity

$$y_{st} = b\, T_s^{[t-5,\, t-1]} + \alpha_{\mathrm{age}_{st}} + \gamma_t + \delta_{\mathrm{SIC}_s} + \varepsilon_{st}$$

	2000–2024	
	log sales$_t$ (1)	log(1+R&D per worker)$_t$ (2)
mean link age $T_s^{[t-5,\, t-1]}$, years	$+0.115^{***}$ (0.012)	-0.031^{***} (0.008)
supplier-years		24,385
suppliers		3,656

T = mean age of the supplier's links over $t-5$ to $t-1$ (median 2.1 years); α_{age}: one dummy per year of the firm's age. Standard errors clustered by supplier. Compustat, US non-financial, non-utility suppliers; 2024 dollars.

◂ Back to This Paper ▸ Appendix: sales-weighted

A Schumpeterian Model with Quality-Adjusted Quantity Fluctuations

Households

- A representative household derives utility from consuming the final good C_t:

$$U = \int_0^\infty e^{-\rho t} \ln C_t \, dt, \qquad \rho > 0$$

- It supplies labour L inelastically at the wage w_t. The wealth evolution equation is

$$\dot{A}_t = r_t A_t + w_t L - C_t$$

Final Good Production

- A continuum of industries, each supplying one intermediate good. Within an industry, competition between firms for the supply of the intermediate goods.
- Production takes two steps at every instant t.
  - Step 1: firms post prices p_{jft}. The final-good producer picks a supplier and orders y_{jft}. The supplier buys labour n_{jft} and input X_{jft} to fill the order.
  - Step 2: the shock $\epsilon_{jft} \sim \mathcal{N}\left(-\frac{1}{2}\sigma_{jft}^2, \sigma_{jft}^2\right)$, $\mathbb{E}\, e^{\epsilon_{jft}} = 1$, is realised, and the supplier delivers

$$\underbrace{e^{\epsilon_{jft}}}_{\text{shock}} \times \underbrace{z_{jft}\, n_{jft}^{1-\alpha} X_{jft}^{\alpha}}_{\text{order } y_{jft}}$$

- A competitive final-good producer combines the intermediate goods with a Cobb–Douglas technology. Shocks are independent, so by the LLN Y_t is not random:

$$Y_t = \exp\left(\int_0^1 \ln\left(e^{\epsilon_{jt}} y_{jt}\right) dj\right) \overset{\text{LLN}}{=} \exp\left(\int_0^1 \left(\ln y_{jt} - \sigma_{jt}^2/2\right) dj\right)$$

▸ Appendix: derivation

- The final good is consumed or used as the input X_{jft} of intermediate firms,

$$Y_t = C_t + \int_0^1 X_{jt}\, dj$$

Effective Price and Limit Pricing within Industries

- Two firms per industry, $f = 1, 2$, each with a posted price p_{jft} and a volatility σ_{jft}. In each industry the final-good producer picks the firm with the lowest effective price, $\tilde{p}_{jft} \equiv p_{jft}\, e^{\sigma_{jft}^2/2}$.
- Market leader limit pricing over the effective price. The leader produces an order at unit cost c_{jt}, the follower at $e^{g_{jt}} c_{jt}$, with g_{jt} the technology difference. The leader's product is new or mature, $s_{jt} \in \{N, M\}$, the follower's is new, and $\sigma_N > \sigma_M$:

$$\underbrace{p_{jt}\, e^{\sigma_{s_{jt}}^2/2}}_{\text{leader's effective price}} = \underbrace{e^{g_{jt}} c_{jt}\, e^{\sigma_N^2/2}}_{\text{follower's effective price at cost}}$$

- The markup and the profit follow, with $\Delta \equiv (\sigma_N^2 - \sigma_M^2)/2$,

$$\mu(g, s) \equiv \frac{p_{jt}}{c_{jt}} = e^{g}\, e^{\Delta \cdot \mathbf{1}[s=M]}, \qquad \pi_t(g, s) = \left(1 - \frac{1}{\mu(g, s)}\right) Y_t$$

Innovation and Successful Taking Over

- Innovation comes with a hazard rate based on R&D effort.
- An innovation builds on the leader's technology z_{jt} and raises it to $z_{jt}\, e^{\lambda}$, with a random step $\lambda \sim \mathrm{Exp}(\theta)$.
- An innovation by the leader, the follower or an entrant succeeds only if its effective price is below the current producer's,

$$\lambda > \Delta \cdot \mathbf{1}[s_{jt} = M]$$
$$p_s \equiv \Pr(\text{adopt} \mid s) = e^{-\theta \Delta \cdot \mathbf{1}[s=M]}$$

Dynamic Problem of Firms: The Leader

- The leader's state is its technology difference from the follower, g, and whether its product is new or mature, $s \in \{N, M\}$.
- A firm becomes the leader only by overtaking the other firm, and its product matures only after that, so $g > 0$.
- The HJB of the leader is

$$
\begin{aligned}
rV_t(g,s) - \dot{V}_t(g,s) = \max_{x_{g,s}} \Bigg\{ & \underbrace{\pi_t(g,s) - c_t(x_{g,s})}_{\text{profit net of R\&D cost}} \\
& + \underbrace{x_{g,s}\, p_s \left[\mathbb{E}^s_\lambda V_t(g+\lambda, N) - V_t(g,s)\right]}_{\text{own successful innovation}} \\
& + \underbrace{(x_{-g,s} + \tau)\, p_s \left[\mathbb{E}^s_\lambda V_t(-\lambda, N) - V_t(g,s)\right]}_{\text{replaced by the follower or an entrant}} \\
& + \underbrace{m \cdot \mathbf{1}[s = N] \left[V_t(g,M) - V_t(g,N)\right]}_{\text{product matures}} \Bigg\}
\end{aligned}
$$

- R&D cost: $c_t(x) = \frac{\psi}{2} x^2 w_t$.

Dynamic Problem of Firms: The Follower

The follower's state is $(-g, s)$, its difference from the leader and the leader's product state. The HJB of the follower is

$$
\begin{aligned}
rV_t(-g,s) - \dot{V}_t(-g,s) = \max_{x_{-g,s}} \Big\{ & \underbrace{-c_t(x_{-g,s})}_{\text{R\&D cost}} \\
& + \underbrace{x_{-g,s}\, p_s \left[\mathbb{E}^s_\lambda V_t(\lambda, N) - V_t(-g,s)\right]}_{\text{own successful innovation}} \\
& + \underbrace{\tau\, p_s \left[0 - V_t(-g,s)\right]}_{\text{exits}} \\
& + \underbrace{x_{g,s}\, p_s \left[\mathbb{E}^s_\lambda V_t(-g-\lambda, N) - V_t(-g,s)\right]}_{\text{the leader innovates successfully}} \\
& + \underbrace{m \cdot \mathbf{1}[s = N] \left[V_t(-g, M) - V_t(-g, N)\right]}_{\text{the leader's product matures}} \Big\}
\end{aligned}
$$

Free Entry

- Entry costs ν research workers per innovation. An innovation lands in a random industry, whose product is of type s with probability ω_s.
- Free entry: the rate of entrants' innovations per industry, τ, adjusts until the cost equals the expected value of an innovation,

$$\nu\, w_t = \sum_{s \in \{N,M\}} \omega_s\, p_s\, \mathbb{E}^s_\lambda V_t(\lambda, N)$$

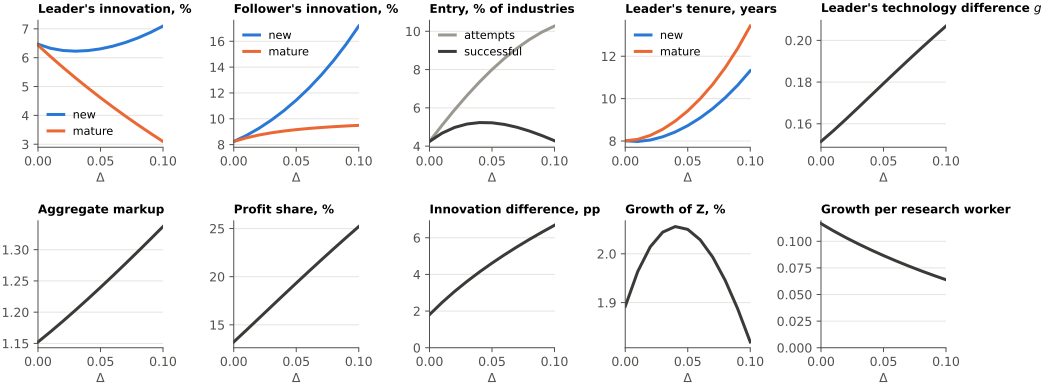
▸ Appendix: balanced growth path, industry structure, aggregation

Quantitative Simulation [Future Work]

Parameters

	Parameter	Value	Set to
ρ	discount rate	0.03	real interest rate 3%
α	input share	0.5	intermediate share of gross output
θ	innovation step $\lambda \sim \mathrm{Exp}(\theta)$	10	mean step 10%
m	maturation rate	1	a new product matures in a year
ψ	R&D cost	14.8	growth of Z 2.05% at $\Delta = 0.05$
ν	entry cost	1.91	entrants' innovation rate 8% per year at $\Delta = 0.05$
Δ	new-product discount	$0 \to 0.10$	the experiment; leader's tenure 8–12 years

Comparative Statics: Δ Increases after 2000



Next Steps

- **Calibration.**
- **Identification of Δ.**
- **Welfare analysis.**

Conclusion

Conclusion

- **Facts.** New links deliver a more volatile supply, and increasingly so since 2000. Links last longer at large, stable suppliers, and have lasted longer since 2000. Suppliers with lasting links are large and R&D-light.
- **Model.** Schumpeterian growth with quality-adjusted quantity fluctuations. Buyers value a stable supply, so a mature leader is harder to take over and has less incentive to innovate.
- **Result.** A rising volatility difference lengthens leaders' tenure, raises markups and slows growth.

Appendix: Known Facts, Numbers and Sources

Entry	
Firm entry rate: 12.0% → 10.6% → below 8%	Decker et al. (2014)
Employment share of firms aged 5 or less: 18.9% → 13.4%	Decker et al. (2014)
Reallocation	
Job creation 18.9% → 15.8%, destruction 16.1% → 13.4%	Decker et al. (2014)
Firm growth less dispersed; fewer high-growth young firms	Decker et al. (2016)
Employment responds less to the firm's own productivity	Decker et al. (2020)
Market structure	
Sales concentration rises	Autor et al. (2020); Grullon, Larkin & Michaely (2019)
Markup 21% → 61%; profit rate 1% → 8%	De Loecker, Eeckhout & Unger (2020)
Labour share falls, most where concentration rises most	Karabarbounis & Neiman (2014); Autor et al. (2020)
Growth	
Labour productivity growth slows after 2004	Syverson (2017)
Research effort rises, research productivity falls	Bloom et al. (2020)

United States. Entry and job flows: average of the late 1980s → average of 2004–06 (→ Great Recession); markup over marginal cost and profit rate: 1980 → 2016. Decker et al. = Decker, Haltiwanger, Jarmin & Miranda. List after Akcigit & Ates (2021).

Back

Appendix: Fact 1, 1981–2024

ratio of SDs of sales growth

	1981–85	1986–90	1991–95	1996–00	2001–05	2006–10	2011–15	2016–20	2021–24
age 1 / age 2	1.16	1.15	1.14	1.13	1.12	1.09	1.22	1.24	1.24
age 1 / age 3	1.34	1.20	1.24	1.20	1.19	1.19	1.33	1.32	1.31
age 1 / age 4	1.60	1.26	1.23	1.26	1.25	1.25	1.33	1.32	1.53
age 1 / age 5	1.50	1.49	1.27	1.35	1.32	1.30	1.42	1.36	1.48
age 1 / age 6	1.42	1.41	1.35	1.48	1.37	1.37	1.46	1.70	1.39
age 1 / age 7	1.83	1.30	1.44	1.31	1.37	1.45	1.55	1.81	1.74
age 1 / age 8+	1.66	1.81	1.74	1.60	1.56	1.62	1.76	1.86	2.12

Back to Fact 1

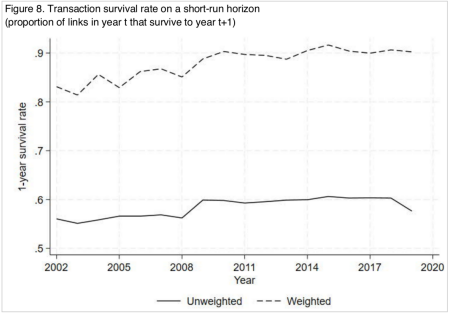
Appendix: Fact 2, Predetermined Characteristics

	baseline, x_{t-1}	supplier alive in $t+5$	x_{t-3}	x_{t-5}
	2000–24	2000–19	2000–24	2000–24
log employment	$+1.72^{***}$	$+1.65^{***}$	$+1.84^{***}$	$+1.52^{***}$
	(0.32)	(0.40)	(0.35)	(0.38)
log(1+R&D per worker)	$+0.26$	$+0.42$	$+0.20$	$+0.08$
	(0.37)	(0.46)	(0.42)	(0.47)
SD of sales growth, five-year window	-2.16^{***}	-2.26^{***}	-1.50^{***}	-1.78^{***}
	(0.31)	(0.44)	(0.34)	(0.37)
link-years	67,228	36,371	51,948	41,508

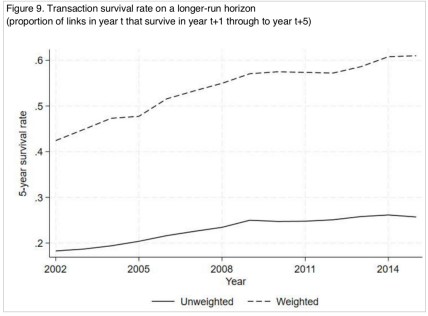
Back to Fact 2

Appendix: Fact 3 in Belgium, All B2B Links

A. 1-year survival rate — NBB WP444 Fig. 8 (2002–2020)



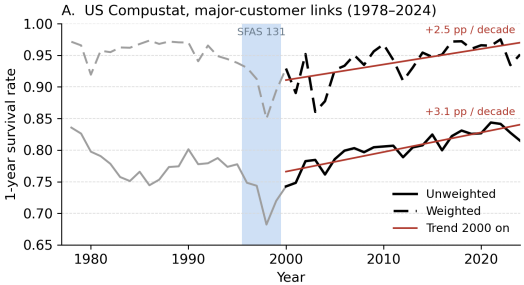
B. 5-year survival rate — NBB WP444 Fig. 9 (2002–2015)

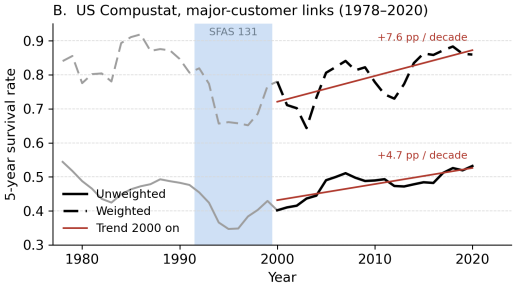


Source: National Bank of Belgium Working Paper 444, Figures 8 and 9. All firm-to-firm links in Belgium, 2002 onward.

Back to Fact 3

Appendix: Fact 3 on the Full Sample, 1978–2024

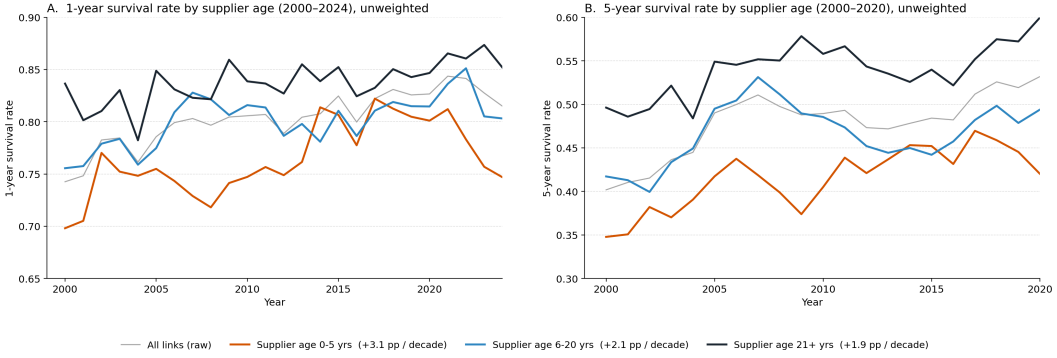




US Compustat, supplier–major-customer links. Survival rate = share of year-t links still reported in $t+1$ (A) or through $t+5$ (B); weighted by the link's share of year-t link sales. Blue: years touched by the 1998 SFAS 131 re-collection. Grey: before 2000. Red: trend from 2000, pp per decade.

Back to Fact 3

Appendix: Fact 3 by Supplier Age



US Compustat, supplier–major-customer links, unweighted. Each line: the survival rate of the year-t links whose supplier has been listed for the stated number of years. Legend: least-squares trend from 2000, pp per decade. Links at suppliers listed 21+ years: 17% of links in 2000–04, 45% in 2020–24.

Back to Fact 3

Appendix: Fact 4 with the Sales-Weighted Link Age

$$y_{st} = b\, T_s^{w,[t-5,\,t-1]} + \alpha_{\mathrm{age}_{st}} + \gamma_t + \delta_{\mathrm{SIC}_s} + \varepsilon_{st}$$

$T_s^{w,[t-5,\,t-1]}$ = the sales-weighted mean age of s's links over $\tau \in [t-5, t-1]$

	2000–2024	
	log sales$_t$	log(1+R&D per worker)$_t$
	(1)	(2)
sales-weighted mean link age $T_s^{w,[t-5,\,t-1]}$, years	$+0.105^{***}$	-0.029^{***}
	(0.011)	(0.007)
supplier-years		22,840
suppliers		3,500

Back to Fact 4

Appendix: Derivation of the Volatility Discount

- Shocks are independent across industries, so the LLN replaces each ϵ_{jt} by its mean:

$$\int_0^1 \ln\left(e^{\epsilon_{jt}} y_{jt}\right) dj = \int_0^1 \left(\mathbb{E}\left[\epsilon_{jt}\right] + \ln y_{jt}\right) dj$$

- The shock is lognormal with $\mathbb{E}\, e^{\epsilon_{jt}} = 1$, so its log has a negative mean:

$$\mathbb{E}\left[\epsilon_{jt}\right] = -\frac{\sigma_{jt}^2}{2} < 0 = \ln \mathbb{E}\, e^{\epsilon_{jt}}$$

- The certainty-equivalent delivery is $e^{-\sigma_{jt}^2/2}\, y_{jt}$, so

$$Y_t = \exp\left(\int_0^1 \left(\ln y_{jt} - \frac{\sigma_{jt}^2}{2}\right) dj\right)$$

- Intuition: the technology is concave in each good, so a shortfall costs more than an overshoot gains. Under CES with elasticity η the discount is $e^{-\sigma_{jt}^2/(2\eta)}$.

Back to Final Good Production

Appendix: Balanced Growth Path

Let $f(g, s)$ be the density of industries over the leader's state. Flow out of each state equals flow in:

$$
\begin{aligned}
(x_{g,N} + x_{-g,N} + \tau + m)\, f(g, N) = &\int_0^g \left[x_{g',N} f(g', N) + x_{g',M} \mathbf{1}[g - g' > \Delta] f(g', M)\right] \theta e^{-\theta(g-g')}\, dg' \\
&+ \theta e^{-\theta g} \int_0^\infty (x_{-g',N} + \tau) f(g', N)\, dg' \\
&+ \theta e^{-\theta(g-\Delta)} \mathbf{1}[g > \Delta]\, p_M \int_0^\infty (x_{-g',M} + \tau) f(g', M)\, dg'
\end{aligned}
$$

$$
(x_{g,M} + x_{-g,M} + \tau)\, p_M\, f(g, M) = m\, f(g, N)
$$

A balanced growth path is a set of values, innovation rates, τ and a density that satisfy the two HJBs, free entry, labour-market clearing and these conditions, with $V_t \propto w_t \propto Y_t$ and $r - g = \rho$.

Back to the model

Appendix: Steady-State Industry Structure

- Shares of industries by the leader's product; average innovation rates of leaders and of followers:

$$\omega_s = \int_0^\infty f(g,s)\,dg, \qquad \bar{x}_s = \frac{1}{\omega_s}\int_0^\infty x_{g,s}\,f(g,s)\,dg, \qquad \bar{x}_s^F = \frac{1}{\omega_s}\int_0^\infty x_{-g,s}\,f(g,s)\,dg$$

- Flow out of the mature state equals flow in:

$$\left(\bar{x}_M + \bar{x}_M^F + \tau\right) p_M\,\omega_M = m\,\omega_N, \qquad \omega_N + \omega_M = 1 \quad\Longrightarrow\quad \omega_M = \frac{m}{m + \left(\bar{x}_M + \bar{x}_M^F + \tau\right) p_M}$$

- Replacement rate and tenure of a leader:

$$\bar{h} = \omega_N(\bar{x}_N^F + \tau) + \omega_M(\bar{x}_M^F + \tau)\,p_M, \qquad \text{tenure} = 1/\bar{h}$$

- Growth: $1/\theta$ per successful step, Δ more when a mature product is replaced, by a challenger or by its own producer:

$$g_Z = \left(\frac{1}{\theta} + \Delta\right)\omega_M\left(\bar{x}_M + \bar{x}_M^F + \tau\right)p_M + \frac{1}{\theta}\,\omega_N\left(\bar{x}_N + \bar{x}_N^F + \tau\right)$$

Parameters of the Δ experiment, $\Delta = 0.05$: $\omega_M = 0.88$, $\bar{x}_N = 6.3\%$, $\bar{x}_M = 4.6\%$, $\bar{x}_N^F = 11.4\%$, $\bar{x}_M^F = 9.2\%$, $\tau = 8.0\%$, tenure 8.7 years, $g_Z = 2.05\%$.

Back to the model

Appendix: Aggregation

Let $\Lambda \equiv \int_0^1 \mu_{jt}^{-1}\, dj$ be the cost share of sales.

The final-good producer spends Y_t on each industry. Firms buy input $X_t = \int_0^1 \alpha\, c_{jt} y_{jt}\, dj = \alpha\Lambda\, Y_t$, so consumption, which is GDP, is

$$C_t = (1 - \alpha\Lambda)\, Y_t$$

Workers produce or do research, $L = L_P + L_R$, and the production labour market clears: $w_t L_P = (1-\alpha)\,\Lambda\, Y_t$. The numeraire condition gives the wage:

$$w_t = \left[\alpha^{\alpha}(1-\alpha)^{1-\alpha}\,\frac{Z_t\,\bar{a}}{\mathcal{M}}\right]^{\frac{1}{1-\alpha}}, \qquad Z_t \equiv e^{\int_0^1 \ln z_{jt}\, dj}, \quad \mathcal{M} \equiv e^{\int_0^1 \ln \mu_{jt}\, dj}, \quad \bar{a} \equiv e^{-\int_0^1 \sigma_{jt}^2/2\, dj}$$

The profit share of sales is $1 - \Lambda$. On a balanced growth path w_t and Y_t grow at $g = g_Z/(1-\alpha)$.

Back to the model